

Building My First Full-Stack Mini App

The Digital Kirana Shop 

 **Welcome, Little Engineers!**

Imagine a wonderful shop where you can buy your daily groceries online, anytime, anywhere! That's the **Digital Kirana Shop**—your very own mini online store for yummy snacks, colorful spices, and everything you love!

In this fun lesson, we are going to open the hood of a digital application and look at the secret clockwork that makes online shopping apps come to life. Let's start our coding adventure together!

What Does "Full-Stack" Mean?

When we build a website or an app on a tablet, computer, or mobile phone, we call it a "**Full-Stack Mini App**" because it is built out of different layers stacked neatly together, just like a delicious club sandwich or a tall tower of toy blocks!

Think Like a Puppet Show!

To understand how an app works, let's imagine we are watching a colorful, magical puppet show:

- **Frontend:** These are the beautiful puppets you see dancing on the stage! You can watch them, touch them, and click on them.
- **Backend:** This is the clever, hardworking puppeteer standing hidden behind the heavy velvet curtain making it all move and work!
- **Database:** This is the sturdy wooden puppet box backstage where all the extra puppets are safely stored and kept organized when they aren't performing!

Meet the Key Parts of Our Kirana Shop!

Now, let's leave the puppet theater and walk right into our online grocery store. Our digital shop has the exact same three parts:



The 3 Building Blocks of Our App

- **1. Frontend (The Shop Window):** The frontend is what you see directly on your tablet screen. It displays bright pictures of crisp snacks, shiny order buttons, and neat pages. Just like choosing your favorite candy right through the clear glass of a real shop window, you tap on what you want!
- **2. Backend (The Shopkeeper's Brain):** The backend is like a very smart, invisible shopkeeper who knows all the store rules and packs your packages. When you click a snack, the backend double-checks if it is ready in the back room!
- **3. Database (The Storeroom):** The database is like a huge, tidy storeroom stacked high with shelves where every single item is safely kept. It always remembers exactly what items are left in stock and counts how many remain!



How Our Kirana Shop Works

Every time you or your friends buy an item using an app, a fast step-by-step process happens behind the scenes inside the computer:

1

Choose Items

You pick your favorite biscuits or spices from the colorful **Frontend** shop window and click buy.

2

Check Stock

The **Backend** shopkeeper quickly runs to the **Database** storeroom to make sure the item is ready.

3

Pack & Deliver

Once found, your order is securely packed up and delivered straight to your application dashboard door!

Steps to Build Your Own Mini App

Are you ready to become a software creator and build your own shop? Here are the milestones we take on our journey:

- 🎨 **Design your shop's look (Frontend):** Create the colorful buttons, item pictures, and menus that your shoppers will tap.
- ⚙️ **Write the shopkeeper's rules (Backend):** Write out the logical rules that tell your store how to count costs and handle orders.
- 📦 **Set up the storeroom (Database):** Save list items like rice, tea, and cookies securely so customers can discover them.

👥 Meet Our Happy Shoppers!

Look at our smart classroom users at work! **Aarav** wants sweet biscuits, **Diya** wants warm tea, **Kabir** wants healthy rice, and **Ananya** helps pack all the orders nicely. Each kid uses the tablet app to shop for their favorite daily treats. By learning how to code, you can build tools that make your friends smile too!



Student Practice Worksheet

Let's test your super developer brain! Answer the questions below using what you learned in this lesson.

1. What are the three main building blocks (parts) of a Full-Stack Mini App?

2. If you are comparing an app to a puppet show, what represents the "Frontend"?

3. What is the job of the "Backend" in our Digital Kirana Shop?

4. Which part of the app acts like a large storeroom that remembers what is in stock?

5. Imagine Aarav clicks on a picture of biscuits on his tablet. Which layer of the app is he interacting with?



6. Why do we call it a "Full-Stack" app? Describe the shape or image it reminds you of.

7. What is Step 2 in the step-by-step process of how our Kirana Shop handles an order?

8. If the database storeroom runs out of tea, who tells the user that it is out of stock?

9. What is the very first task you need to do when designing the "Frontend" look of your app?

10. True or False: The backend puppeteer stands out on the front stage where everyone can see them.

11. Name three items mentioned in the lesson that are stored inside our Kirana Shop database.

12. Once you learn about databases, what exciting new thing will you be able to do in your app?

13. Draw a line or match the puppet show terms with app parts: Frontend is the _____, Backend is the _____, and Database is the _____.

🌟 Fantastic Job! You are ready to build the future! 🌟